# Which tag pair should an indie game use? First-Person × Multiplayer

MADT 7202 Mid-Term · Panjapol Ampornratana 6810424009 · Repo: github.com/hhej/steam-tag-whitespace

Steam is the largest PC game store. Developers set a game's tags, players vote on them; top tags drive search.

## 1 The decision

An indie studio's creative director picks the next game's two lead tags; if no pair passed the selection rule (section 5), they would pick a crowded pair (table).

## 2 Why a network

Counts show how common a pair is and how well it sells, not whether its tags belong to separate groups or which tag connects them. That needs the network.

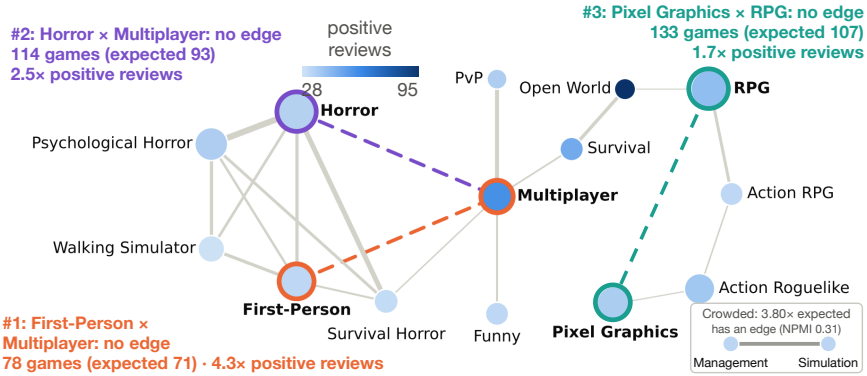
## 3 Data

| | |
|---|---|
| **Source** | S1 FronkonGames steam-games-dataset (Hugging Face); S2 Steam store. |
| **Obtained** | S1 Sep 2025 snapshot. S2: 1,000 random S1 games (seed 7202), scraped 2026-10-14: tags, review counts. |
| **Legal** | S1 MIT licence; S2 allowed by robots.txt, 1 request/1.5 s; no personal data. |
| **Volume** | S1: 9,094 Indie games (2023–24, ≥10 reviews) + 2,076 from Jan–Jun 2025 as a test set. S2: 996. |
| **Check** | S1 and S2 agree: 87% top-5 tag overlap; reviews correlate 0.98. |
| **Bias** | Favours games that sold; reviews count buyers, not demand; late-2024 games had less time. |

## 4 Graph construction

| | |
|---|---|
| **Node** | A Steam tag. Undirected, weighted tag–tag projection of the game–tag bipartite graph. |
| **Edge** | Co-occurrence in a game's top 5 tags; suggests relatedness, not a stated relation. |
| **Resolution** | Co-op, Online/Local Co-Op and Local Multiplayer merged into Multiplayer (84% of Horror × Multiplayer games were also tagged co-op). |
| **Weight** | NPMI (co-occurrence above expected). Raw counts give 9 of the top 10 edges to the 5 most common tags; NPMI gives none. |
| **Window** | Top 5 tags per game, as the store shows; with all 20, #1 falls from 4.3× to 2.7× positive reviews. |
| **Excluded** | "Indie", Early Access, Free to Play, adult; on >25% of games: Singleplayer, Adventure, Action, Casual, 2D, 3D, Atmospheric, Exploration. |
| **Pruning** | Tag on ≥1.5% of games (136); edge needs NPMI ≥ 0.10 and ≥0.5% of games (45) with both tags. 439 tags → **80 nodes, 177 edges**. |

## 5 The picture




**First-Person and Multiplayer have no direct link; the shortest route between them goes through Survival Horror (2 links). Without it, the shortest route is 5 links.** 14 of 80 tags; size = games; line width = NPMI.

*Expected* = games with both tags if picked independently (A's games × B's games ÷ 9,094); *crowding* = games ÷ expected; *positive reviews* = thumbs-up reviews of the pair's median game. **Rule:** of pairs on ≥73 games (0.8%, for stable medians), keep crowding ≤ 1.5 and ≤20% of combined neighbours shared; rank by positive reviews.

| Rank: pair | Games | Expected | Crowding | Shared neighbours | Positive reviews | % positive |
|---|---|---|---|---|---|---|
| #1: **First-Person × Multiplayer** | 78 | 71 | 1.10× | 8% | 148 (4.3×) | 76% |
| #2: **Horror × Multiplayer** | 114 | 93 | 1.23× | 8% | 85 (2.5×) | 73% |
| #3: **Pixel Graphics × RPG** | 133 | 107 | 1.25× | 0% | 58 (1.7×) | 88% |
| Crowded (box): **Management × Simulation** | 121 | 32 | 3.80× | 8% | 111 (3.3×) | 79% |
| ***Typical indie game*** | | | | | *34 (1.0×)* | *87%* |

## 6 Findings

- **#1**: 78 games vs 71 expected (NPMI 0.02, under the 0.10 edge cutoff), yet 4.3× typical positive reviews (permutation test, $p < 0.001$); horror games overall 1.0× typical. An association, not cause; reach, not rating (76% positive).
- **Survival Horror** is the only tag linking #1's two neighbour groups; if it spreads, the gap closes.
- **Narrow lead:** top at all 8 pruning settings (4 NPMI cutoffs × 2 minimums) but in 60 of 100 resamples; 2025 games: crowding 1.14, 2.2× positive reviews.

## 7 Recommended action

- Week 0: pitch the game.
- Weeks 1–6: prototype; playtest target 87% positive.
- Week 7: Coming Soon page with both tags.
- Quarterly: re-check crowding.

## 8 What would change my mind

If #1's crowding passes 1.5 or Survival Horror's share of releases rises, I switch to #2 (1.45 in 2025), then #3; if playtests miss 87%, to #3 (88% positive).